Question of interest: ***Are movement signals in V1 the same kind of signal as movement signals in cerebellum, or do they just look similar at the decoding level?***

Stringer et al. (2019) and the IBL Brain-Wide Map (IBL et al., 2025) both report that locomotion and orofacial movement variables are decodable from neurons essentially everywhere in the mouse brain. Stringer et al. analyzed spontaneous neuronal firing, finding that neurons in the primary visual cortex encoded both visual information and motor activity related to facial movements, and the IBL probes covered 279 brain areas in the left forebrain and midbrain and the right hindbrain and cerebellum, with movement-correlated activity appearing across that full coverage.

The question this project takes on is whether that ubiquity is one signal or many. A V1 neuron firing during running and a cerebellar neuron firing during running are both "movement-correlated", but they almost certainly reflect different computations. V1's signal is some mix of efference copy, arousal-driven gain, and retinal slip. Cerebellum's is forward-model prediction error and proprioceptive integration. M1's is the command itself. Hippocampus's is locomotion-modulated theta gating. So is the brain-wide movement signature an inherited copy of one global low-dimensional state, or are these regions running their own distinct movement-related computations that just happen to correlate at the population level?

Decoding accuracy alone cannot answer this, because two regions can reach high decoding scores by having similar information in completely different representational forms. The project therefore needs methods that test for *shared subspace structure* across regions, not just shared *information*.

Dataset: IBL Brain-Wide Map (2025 release)

It is the only public dataset with simultaneous Neuropixels coverage of **V1 (VISp)**, **cerebellum (CB cortex and nuclei)**, **motor cortex (MOp, MOs)**, and **hippocampus (CA1, DG, etc.)** under a ***single standardized task across labs***. 621,733 neurons recorded with 699 Neuropixels probes across 139 mice in 12 laboratories, with the task containing sensory, motor, and cognitive components.

- The Stringer 2019 8-probe Neuropixels dataset is publicly available and is conceptually the origin of this question, but **its probes target forebrain regions and do not cover cerebellum**, so it cannot answer the V1-vs-CB version of the question. Use it as a sanity-check secondary dataset for the V1 side only if you want to validate that your V1 results match Stringer 2019.

For the project, a tractable subset is something like: 30-60 sessions where at least one probe has **VISp units** and at least one probe has **CB units** (or one of CB cortex / cerebellar nuclei / fastigial / interpositus). We can also pull **MOp/MOs** and **CA1** from the same sessions to anchor the comparison with regions whose movement-signal computational identity is much better established (M1 = command, hippocampus = locomotion-gated theta).

Literature review: methods used to study brain-wide movement signals

There are roughly five method families in the relevant literature, and our project will likely need to combine two or three of them.

Per-region encoding GLMs with movement kernels

The dominant brain-wide approach, fit a Poisson GLM per neuron with separate kernels for each behavioral variable (stimulus onset, choice, wheel velocity, pupil, body motion energy, individual DLC pose markers), then attribute variance to each kernel via ΔR^2. This is what Musall et al. (2019) and the IBL BWM paper itself use. Musall et al. found that cortex-wide activity was dominated by movements, especially uninstructed movements not required for the task, and accounting for them revealed that neurons with similar trial-averaged activity often reflected utterly different combinations of cognitive and movement variables. The IBL paper uses ΔR^2 from movement and stimulus kernels to map variance explained per region (PubMedPubMed)

- Strengths: interpretable, per-neuron, well established.
- Weakness: variance attributed to movement kernels in V1 and CB will both be nonzero, but the GLM does not say whether that variance is *the same kind* of signal. ΔR^2 ranks regions, it does not separate computational role.

The most relevant recent paper is **Wang, Kurgyis et al., Nat Neurosci 2026** ("Brain-wide analysis reveals movement encoding structured across and within brain areas"). They used multiple machine learning methods to predict neural activity from videography and found that movement-related signals differed across areas, with stronger movement signals close to the motor periphery and in motor-associated subregions, and delineating activity that predicts or follows movement revealed fine-scale structure of sensory and motor encoding. This is the closest published precedent for what we are doing. **They essentially answer "movement signals are not uniform across the brain, they are structured"**, but they do not directly address the V1-vs-cerebellum semantic question. Their predicts-vs-follows timing trick is gold for us (see 3.5 below) (Nature).

Decoding from population activity

Stringer 2019 set the template: **predict facial motion energy SVDs from neural activity, or vice versa, using reduced-rank regression with cross-validation**. Stringer et al. recorded more than 10,000 neurons in mouse visual cortex and found that spontaneous activity reliably encoded a high-dimensional latent state partially related to ongoing behavior. **Sensory inputs did not interrupt this ongoing signal but added onto it a representation of external stimuli in orthogonal dimensions**. The 16-dimensional facemap embedding from Syeda et al. (2024) updates this with a more expressive movement representation (ScienceSteinmetzlab)

For our project, decoding gives us a baseline ranking of how much movement information lives in each region's population, but the result we actually care about is whether the *axes* carrying that information align across regions or not.

Cross-region subspace methods

This is where the methodology has matured fastest in the last few years. The basic idea

- **If V1 and cerebellum carry the same movement signal, there should exist a shared low-dimensional subspace such that the V1 projection onto it and the CB projection onto it co-fluctuate over time**.
- If the signals are computationally distinct, the shared subspace will be small or trivial, and most movement-encoding variance will live in region-specific subspaces.

Tools:

- **Reduced-rank regression (RRR).**
    - Fit Y_CB ~ B X_V1 where rank(B) = r is small. RRR remains a powerful and practical tool for studying interactions between neural populations because it is simple, interpretable, computationally efficient, and often captures a **substantial proportion of shared variance between brain regions**.
    - The cross-validated rank tells us the dimensionality of the shared signal. Semedo et al. (2019) introduced this as the "*communication subspace*" framework (arXiv).
- **Canonical correlation analysis (CCA).**
    - Find **linear combinations in V1 and CB that maximize correlation**. CCA conceptually finds pairs of projection axes from the two populations that maximize the correlation between the projections. Tells you how many shared dimensions exist and how aligned they are (Nature).
- **Partial CCA (pCCA).**
    - Given multivariate variables X, Y and Z, pCCA aims to **seek a shared subspace between X and Y by finding sets of weights that maximize the Pearson correlation coefficient between the weighted X and weighted Y after removing the linear effects of Z**.
    - After partialling out global arousal (pupil, running speed), how much V1-CB shared movement variance remains? If most disappears, the V1 and CB signals were indeed inherited from a common low-d state. If a lot survives, they share movement-related structure beyond global arousal (bioRxiv).
- **Multiplexed subspaces / shared subspace networks.**
    - MacDowell et al. (2023) combined cortex-wide calcium imaging with high-density electrophysiological recordings in eight cortical and subcortical regions of mice and found that different dimensions within the population activity of each brain region were functionally connected with different cortex-wide subspace networks of regions, and these subspace networks were multiplexed, allowing a brain region to simultaneously interact with multiple independent, yet overlapping, networks. This is the methodological state of the art for "is this signal global or

region-specific" questions, and the dimensions-as-channels framing maps directly onto your hypothesis (PubMed Central).

Demixed methods (dPCA, dSCA)

dPCA (Kobak, Brendel, Constantinidis, Feierstein, Kepecs, Mainen, Qi, Romo, Uchida, Machens, 2016) is a single-region method. Demixed principal component analysis **decomposes population activity into a few demixed components that capture most of the variance and highlight dynamic tuning of the population to various task parameters such as stimuli, decisions, rewards**. The multi-region extension is dSCA (Pang & Sahani, 2020). Demixed shared component analysis decomposes population activity into a few components, **such that the shared components capture the maximum amount of shared information across brain regions** while also depending on relevant task parameters, yielding interpretable components that express which variables are shared between different brain regions and when this information is shared across time (arxivarxiv).

- dSCA is the most direct match to the literal phrasing of our question: it explicitly **asks which behavioral variables are shared across V1 and cerebellum versus region-specific, while demixing the contributions of stimulus, choice, and movement**. If dSCA recovers a movement component shared between V1 and CB, that is evidence for a global movement signal. If movement components are mostly region-private, that argues for separate computations.

Suggested pipeline

Course-project-scoped, so don't try to do all five method families. A defensible plan in roughly the order of execution:

1. **Data prep.** Use `bwm_query` to find sessions with simultaneous coverage of VISp + CB (cortex or nuclei) + MOp/MOs + CA1. Filter by good QC units. Bin spikes at 25 ms or 50 ms for the analyses below. Construct movement covariates: wheel velocity and acceleration, body motion energy SVDs from the body camera, DLC pose for paws and snout, pupil diameter, lick. Also keep stimulus onset and choice for control.
2. **Reproduce the per-region movement encoding result.** Fit a kernel-based GLM per neuron with separate movement, stimulus, and choice kernels. Compute ΔR^2 for each kernel per region. We should recover the qualitative finding from IBL BWM and Wang/Druckmann that V1, M1, CB, and hippocampus all carry significant movement variance, with stronger encoding closer to the motor periphery. This is our sanity check.
3. **Within-region SVCA (Stringer) to extract reliable movement-related dimensions per region.** This gives us a denoised low-dimensional movement-related subspace per region, sized by the data, not arbitrary.
4. **Pairwise cross-region analysis on the movement-related subspaces.** Run CCA between V1's movement subspace and CB's, and separately V1-M1, V1-CA1, CB-M1, CB-CA1, M1-CA1. Report the cross-validated canonical correlations and the

dimensionality of the shared subspace. Run pCCA partialling out wheel speed and pupil to ask whether shared structure survives after global state is regressed out. The answer to your question lives almost entirely in this analysis.

5. **dSCA across all four regions jointly.** Demix into stimulus, choice, and movement components, with explicit shared-vs-region-private decomposition. Quantify the fraction of movement variance that is captured by shared components versus region-private components.

Priors: what is known about each region's movement signal

Useful for setting up our discussion section and ensuring that we align with current literature:

- **V1.** Niell and Stryker (2010) showed that neurons in primary visual cortex of awake mice doubled visually-evoked firing rate as the animal transitioned from standing still to running, without changes in spontaneous firing or stimulus selectivity. Locomotion improves encoding of visual stimuli in V1 by both increasing firing rates in upper layers and reducing noise correlations in layer V. The V1 movement signal is best characterized as gain modulation, not feature encoding (PubMed CentralJournal of Neuroscience).
- **Cerebellum.** Strong forward-model and efference-copy theory. A single Purkinje cell can integrate both the efference copy from the motor cortex and sensory feedback from the periphery, forming an adequate substrate for a forward model. In mouse cerebellum, all three classes of cerebellar cells were able to decode the animal's locomotion speed, suggesting the speed signal pervades most computational steps in the cerebellum, consistent with forward internal models receiving efference copy (FrontiersPubMed Central).
- **M1 (MOp/MOs).** Movement command and pre-movement preparatory activity, well established from primate motor cortex literature.
- **Hippocampus.** Theta-locked locomotion modulation, place cells gated by running speed, qualitatively different from any of the above.

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